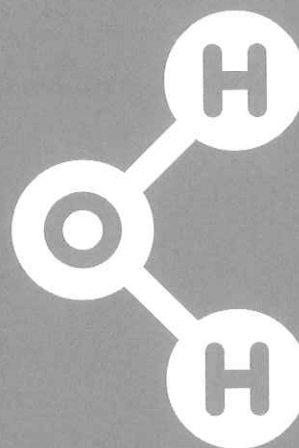




UNSW Global
THE UNIVERSITY OF NEW SOUTH WALES
SYDNEY • AUSTRALIA

PAPER C



2011 ICAS

International Competitions
and Assessments for Schools

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

40 QUESTIONS

TIME ALLOWED: 55 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a 2B or B pencil.

Do **NOT** use a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.
Your score will be the number of correct answers.
Marks are **NOT** deducted for incorrect answers.

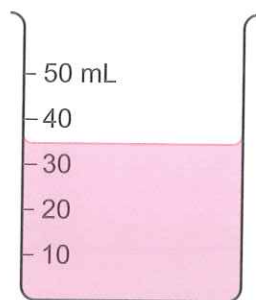
There are **40 MULTIPLE-CHOICE QUESTIONS** (1–40).
Use the information provided to choose the **BEST** answer from
the four possible options.
On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a calculator and a ruler.

SCIENCE

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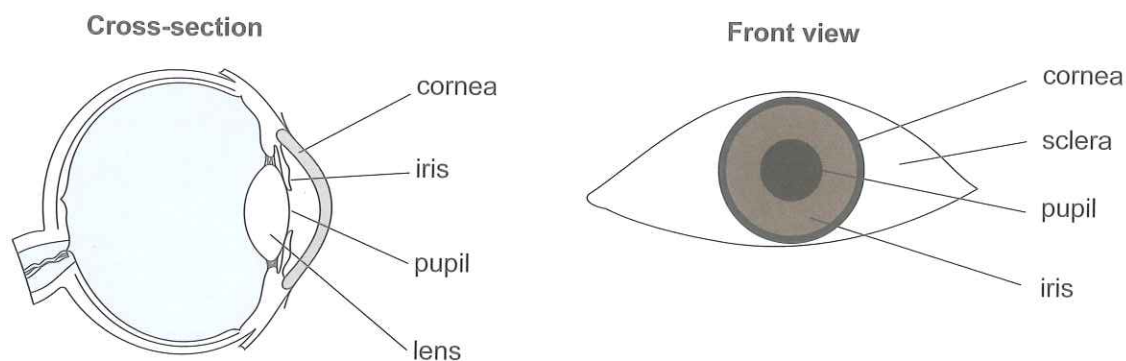
1. This beaker contains some liquid.



What is the volume of liquid in this beaker?

- (A) 25 mL
- (B) 30 mL
- (C) 35 mL
- (D) 40 mL

2. The diagrams show a cross-section and the front view of a human eye.



Which part of the eye is labelled in the cross-section but not in the front view?

- (A) cornea
- (B) iris
- (C) pupil
- (D) lens

3. This table gives the temperature at which some metals become liquid (melting point).

Metal	Melting point ($^{\circ}\text{C}$)
silver	961
iron	1535
gold	1064
copper	1083

Which metal has the highest melting point?

- (A) silver
- (B) iron
- (C) gold
- (D) copper

- Rub out all mistakes completely.
- Print your details clearly in the boxes provided.
- Make sure you fill in only one oval in each column.

FIRST NAME

D	E	B	B	I	E
A	A	A	A	A	A
B	B			B	B
C	C	C	C	C	C
D	D	D	D	D	D
E	E	E	E	E	E

LAST NAME

B	A	C	H
A		A	A
	B	B	E
C	C		
D	D	D	D
E	E	E	E

FIRST NAME **LAST NAME**

C	H	A	N
A	A	A	A
B	B	B	B
C	C	C	C
D	D	D	D
E	E	E	E

A	I		B	E	N	G
A	A	A	A	A	A	A
B	B	B	B	B	B	B
C	C	C	C	C	C	C
D	D	D	D	D	D	D
E	E	E	E	E	E	E

FIRST NAME

J	A	M	A	L		B	I	N
A		A		A	A	A	A	A
B	B	B	B	B	B	B	B	B
C	C	C	C	C	C	C	C	C
D	D	D	D	D	D	D	D	D
E	E	E	E	E	E	E	E	E

LAST NAME

A	B	A	S
A		A	A
B		B	B
C		C	C
D		D	D
E		E	E

[illegible][illegible]

☐ Male ☐ Female



☐ Yes ☐ No

Town / suburb: _____

Today's date: / / Postcode:

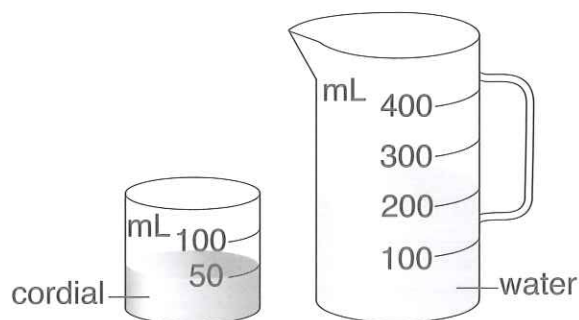
Day		Month		Year	
(0)	(0)	(0)	(0)	(0)	(0)
(1)	(1)	(1)	(1)	(1)	(1)
(2)	(2)		(2)	(2)	(2)
(3)	(3)		(3)	(3)	(3)
(4)			(4)	(4)	(4)
(5)			(5)	(5)	(5)
(6)			(6)	(6)	(6)
(7)			(7)	(7)	(7)
(8)			(8)	(8)	(8)
(9)			(9)	(9)	(9)

(A)	(K)
(B)	(L)
(C)	(M)
(D)	(N)
(E)	(O)
(F)	(P)
(G)	(Q)
(H)	(R)
(I)	(S)
(J)	(T)

TO ANSWER THE QUESTIONS

Example: Ari added cordial to water to make a jug of drink.
What will be the volume of the drink in the jug?

- (A) 50 mL
- (B) 150 mL
- (C) 200 mL
- (D) 250 mL



The answer is 250 mL, so you would fill in the oval **(D)**, as shown.

(A) (B) (C) **(D)**



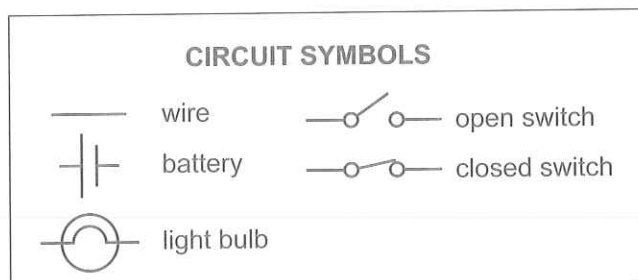
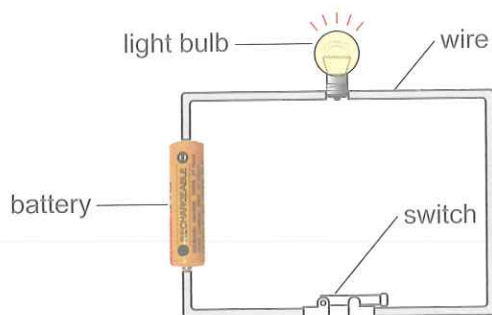
START

- 1 (A) (B) (C) (D)
- 2 (A) (B) (C) (D)
- 3 (A) (B) (C) (D)
- 4 (A) (B) (C) (D)
- 5 (A) (B) (C) (D)
- 6 (A) (B) (C) (D)
- 7 (A) (B) (C) (D)
- 8 (A) (B) (C) (D)
- 9 (A) (B) (C) (D)
- 10 (A) (B) (C) (D)
- 11 (A) (B) (C) (D)
- 12 (A) (B) (C) (D)
- 13 (A) (B) (C) (D)
- 14 (A) (B) (C) (D)
- 15 (A) (B) (C) (D)

- 16 (A) (B) (C) (D)
- 17 (A) (B) (C) (D)
- 18 (A) (B) (C) (D)
- 19 (A) (B) (C) (D)
- 20 (A) (B) (C) (D)
- 21 (A) (B) (C) (D)
- 22 (A) (B) (C) (D)
- 23 (A) (B) (C) (D)
- 24 (A) (B) (C) (D)
- 25 (A) (B) (C) (D)
- 26 (A) (B) (C) (D)
- 27 (A) (B) (C) (D)
- 28 (A) (B) (C) (D)
- 29 (A) (B) (C) (D)
- 30 (A) (B) (C) (D)

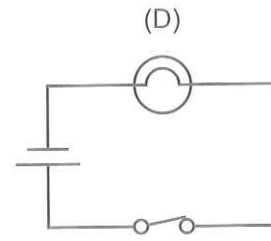
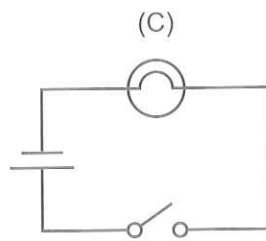
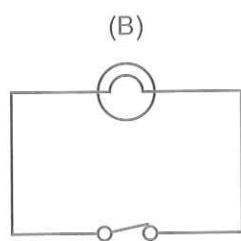
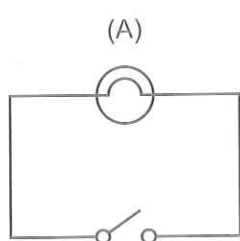
- 31 (A) (B) (C) (D)
- 32 (A) (B) (C) (D)
- 33 (A) (B) (C) (D)
- 34 (A) (B) (C) (D)
- 35 (A) (B) (C) (D)
- 36 (A) (B) (C) (D)
- 37 (A) (B) (C) (D)
- 38 (A) (B) (C) (D)
- 39 (A) (B) (C) (D)
- 40 (A) (B) (C) (D)

4. The diagram shows an electrical circuit and some circuit symbols.

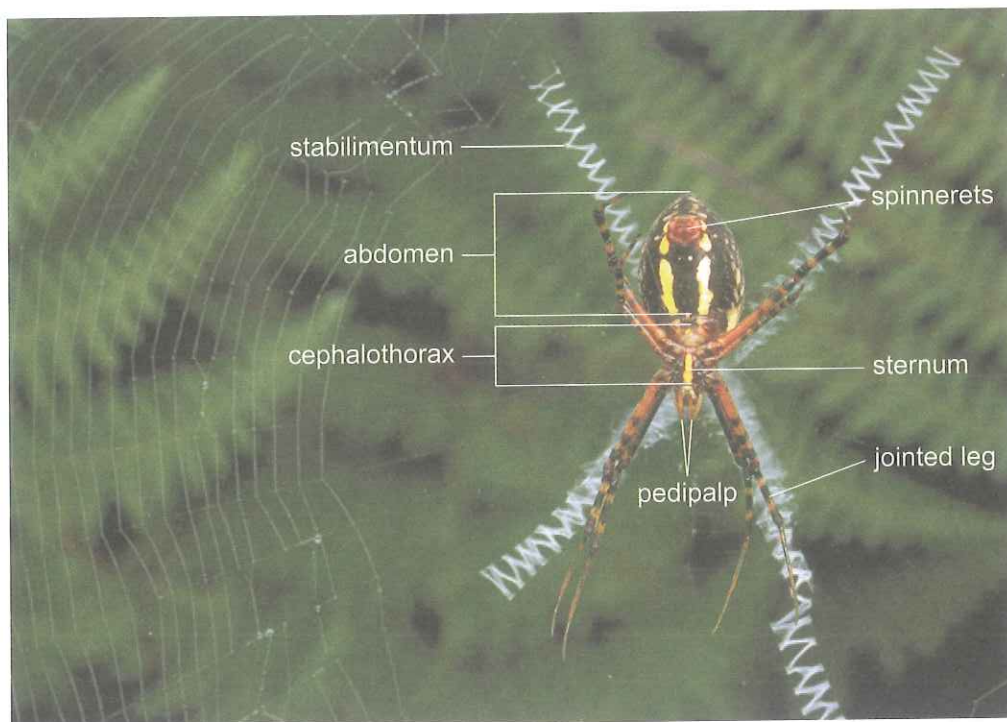


This electrical circuit can also be drawn using the circuit symbols.

Which option shows the electrical circuit drawn using the circuit symbols?



5. The picture shows a spider sitting on its web.

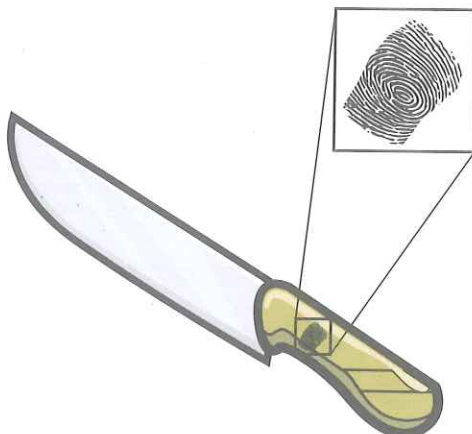


The stabilimentum is

- (A) part of the spider's web.
- (B) part of the spider's abdomen.
- (C) attached to the cephalothorax.
- (D) jointed and attached to the sternum.

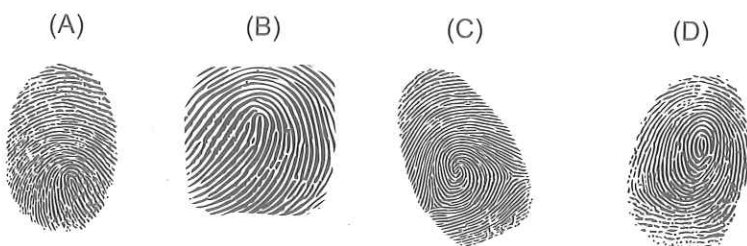
6. A fingerprint is an impression of ridge patterns on a person's finger. Each person has their own unique pattern.

Crime scene investigators found a partial fingerprint impression on a knife used at a crime.



The fingerprints of the suspects are shown below.

Which suspect used the knife?



7. Mandy tested four different surf skis and rated them according to speed (how fast they can go) and manoeuvrability (how easily they can turn).

Mandy will be taking part in a long distance race where she will need a surf ski that will go as fast as possible in a straight line.

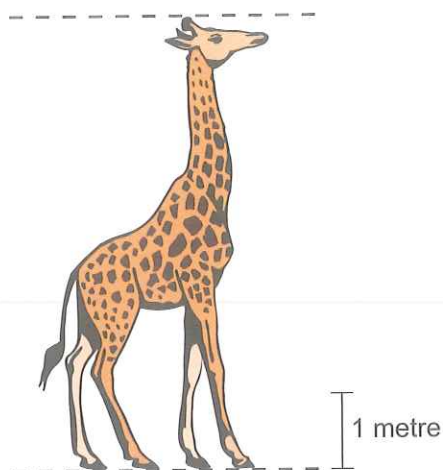
Which surf ski would best suit Mandy's needs?

	Surf ski	Speed rating (1 = fastest)	Manoeuvrability rating (1 = easiest to turn)
(A)		1	4
(B)		3	2
(C)		2	3
(D)		4	1

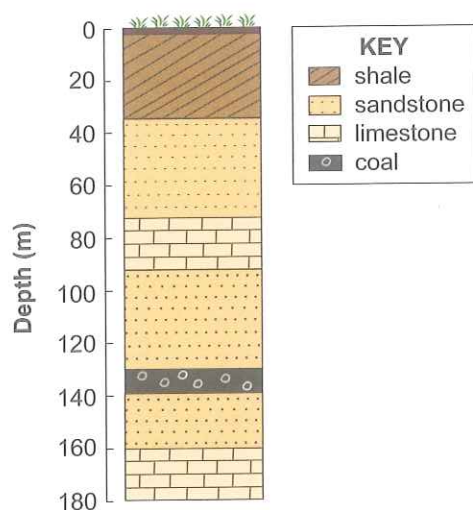
8. This is a picture of a giraffe.

Approximately how tall is the giraffe?

- (A) 3 metres
- (B) 4 metres
- (C) 5 metres
- (D) 6 metres



9. The diagram shows the type of rock layers found under the ground in an area.



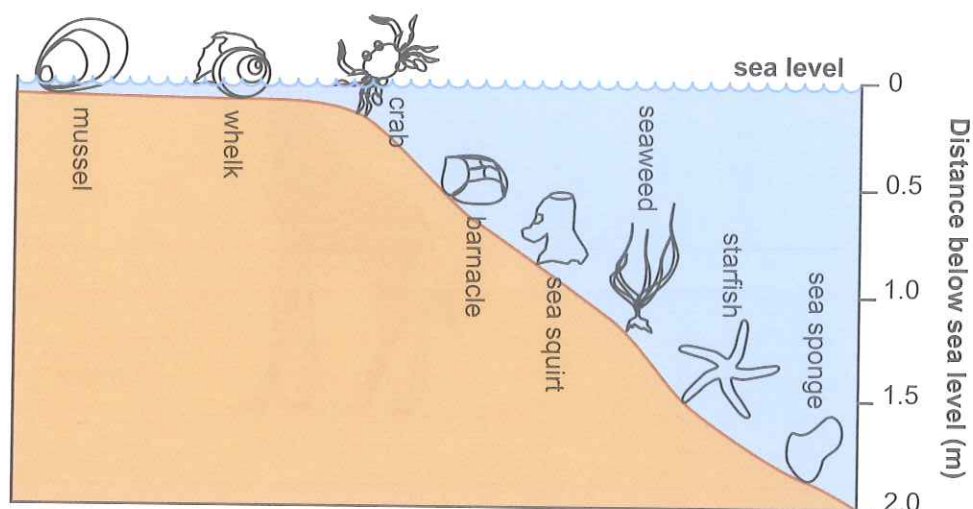
What can be concluded about the rock in this area?

- (A) No limestone formed in this area.
 - (B) Coal formed directly between two layers of limestone rock.
 - (C) Coal is buried under approximately 130 m of rock in this area.
 - (D) Shale can be found at a depth of approximately 100 m in this area.
10. The flakes of some breakfast cereals contain iron filings.

Which would be the best way to separate the iron filings from the cereal?

- (A) use a magnet
- (B) use a magnifying glass
- (C) crush the flakes and then use a magnet
- (D) crush the flakes and then use a magnifying glass

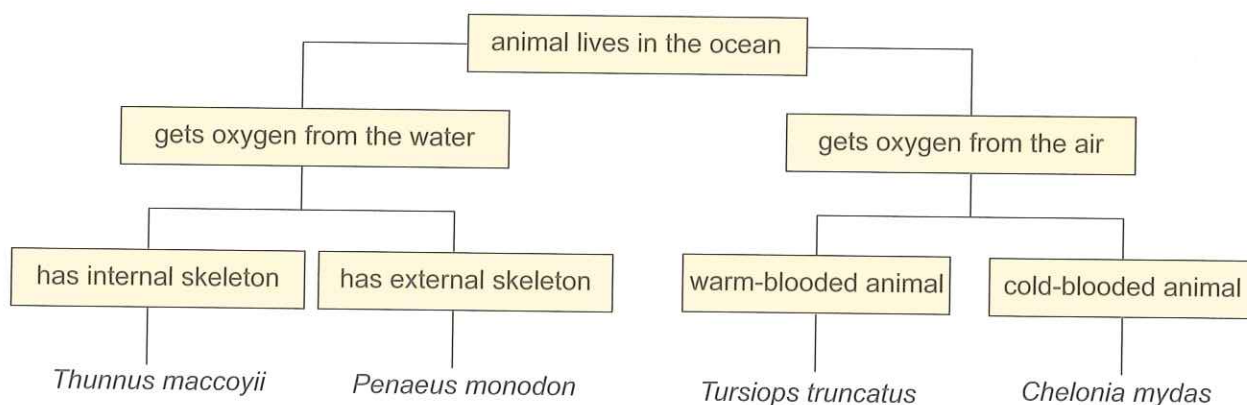
11. The diagram shows the depth at which different organisms were found on a reef.



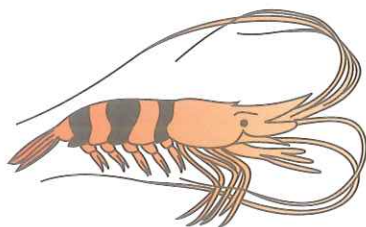
Which organisms were found more than 1 m below sea level?

- (A) mussel and whelk
- (B) crab and barnacle
- (C) sea squirt and seaweed
- (D) starfish and sea sponge

12. The diagram shows information about four animal species that live in the ocean.



A tiger prawn lives in the ocean, gets its oxygen from the water and has an external skeleton.

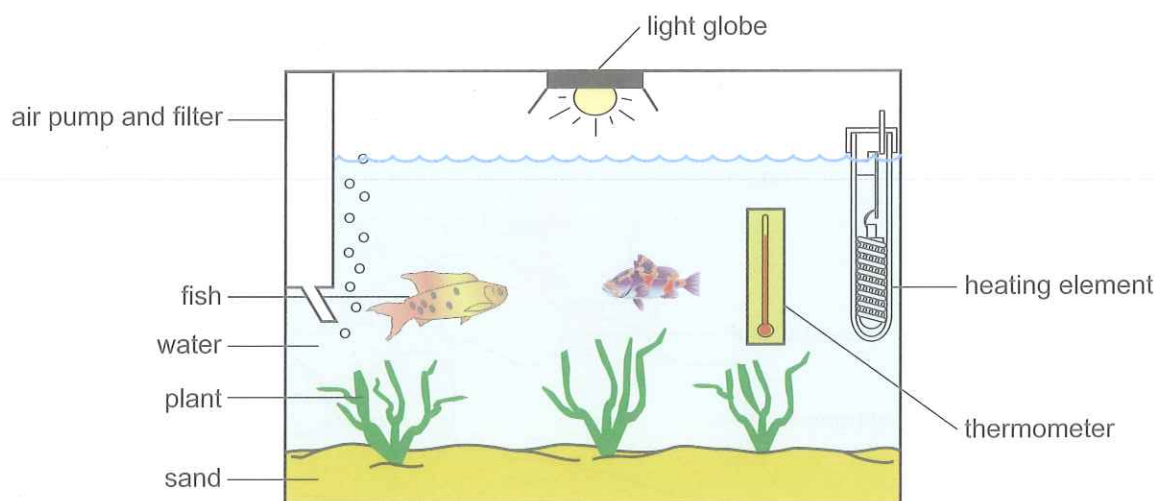


What species is the tiger prawn?

- (A) *Thunnus maccoyii*
- (B) *Penaeus monodon*
- (C) *Tursiops truncatus*
- (D) *Chelonia mydas*

13. Sarah thinks that the plants in her fish tank would grow more quickly if the water in the tank was warmer.

The diagram shows Sarah's fish tank.



What should Sarah do to her fish tank to test her idea?

- (A) change the type of plants used
 - (B) change the type of thermometer used
 - (C) increase the number of fish in the tank
 - (D) increase the time the heating element is on
14. The eye has special cells, called cone cells, that are stimulated by coloured light. There are three types of cone cells: red, blue and green.

Different types of cone cells are stimulated by different colours of light. The table shows the colours of light that stimulate each type of cone cell.

Colour of light detected	Red cone cells	Blue cone cells	Green cone cells
Red	✓		
Blue		✓	
Green			✓
Cyan		✓	✓
White	✓	✓	✓
Yellow	✓		✓
Magenta	✓	✓	

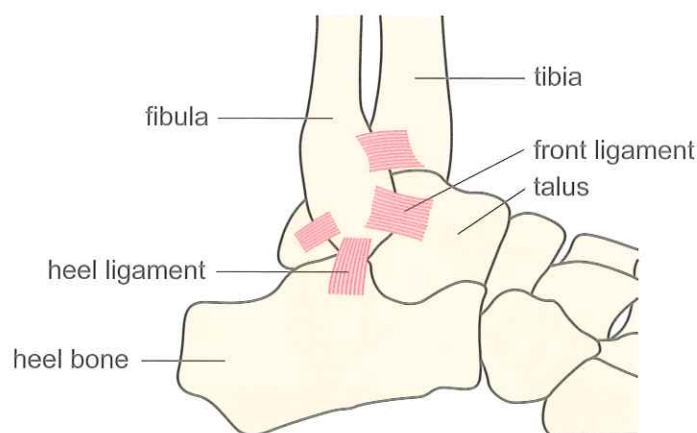
Which cone cells are stimulated by the colour cyan?

- (A) blue and red cone cells
- (B) red and green cone cells
- (C) blue and green cone cells
- (D) red, blue and green cone cells

For questions 15 and 16 use the information below.

Ligaments hold bones together.

The diagram shows some ligaments and bones of the ankle.



15. Which two bones are connected by the front ligament?

- (A) fibula and talus
(B) tibia and fibula
(C) fibula and heel bone
(D) tibia and heel bone

16. The table shows how ankle sprains are graded.

Grade of ankle sprain	Damage to ligament	Symptoms
1	stretched	pain and swelling; can walk but may not be able to jog or jump
2	partly torn	pain and swelling; difficulty walking more than a few steps
3	completely torn	pain, swelling and bruising; finds walking very difficult

Alex sprained her ankle. She has difficulty walking.

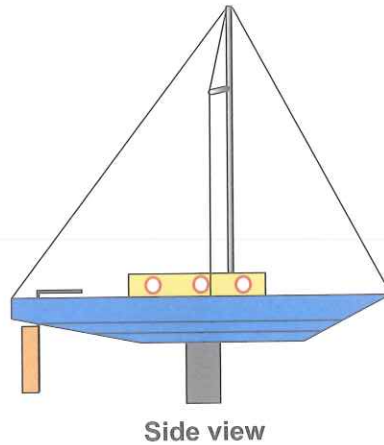
The photograph shows her ankle after she sprained it.

What grade of ankle sprain does Alex have, and which ligament has she most likely damaged?

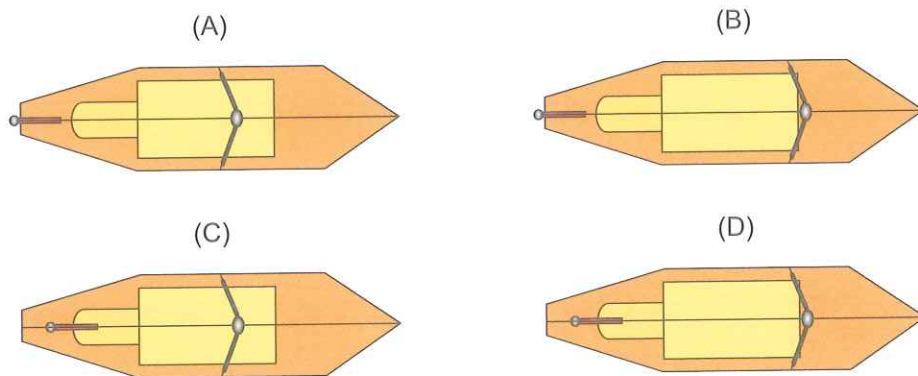


	Grade of ankle sprain	Ligament damaged
(A)	2	heel
(B)	2	front
(C)	3	heel
(D)	3	front

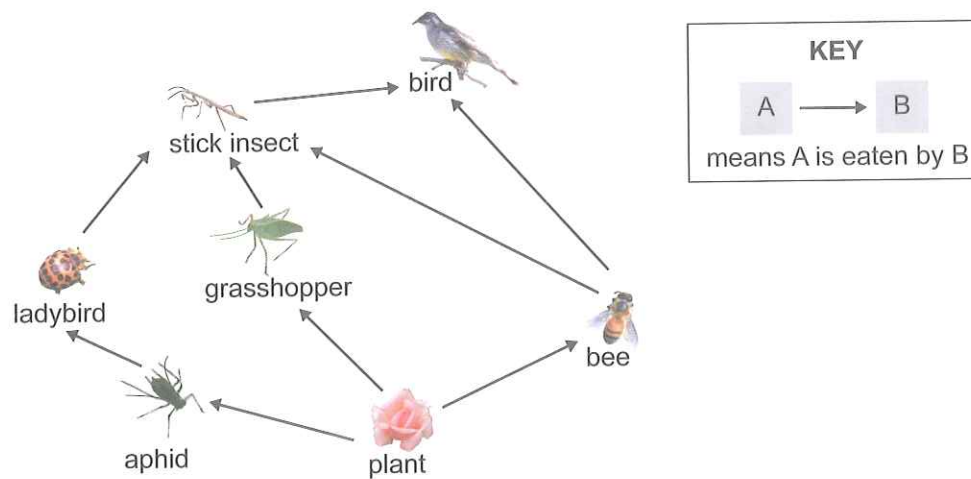
17. The diagram shows a side view of a model boat.



Which diagram shows the boat as seen from above?



18. This is an example of a food web.

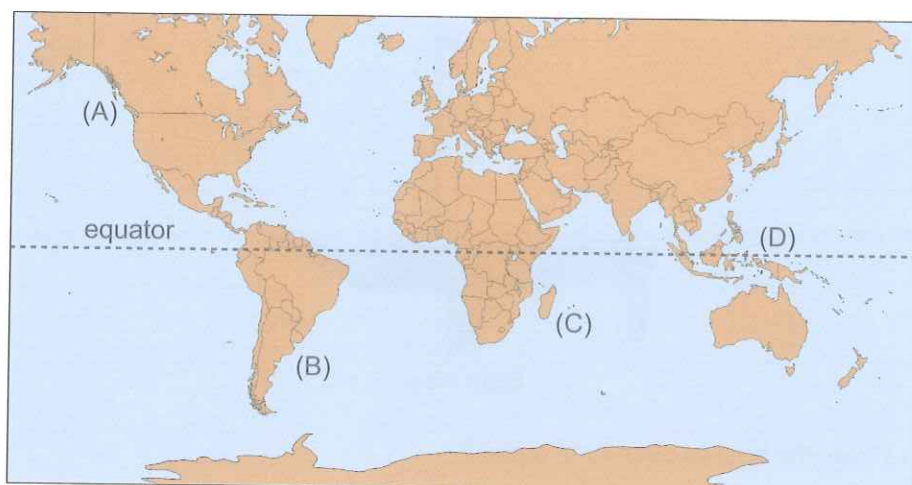


Which animal is **NOT** eaten by the stick insect?

- (A) bee
- (B) bird
- (C) grasshopper
- (D) ladybird

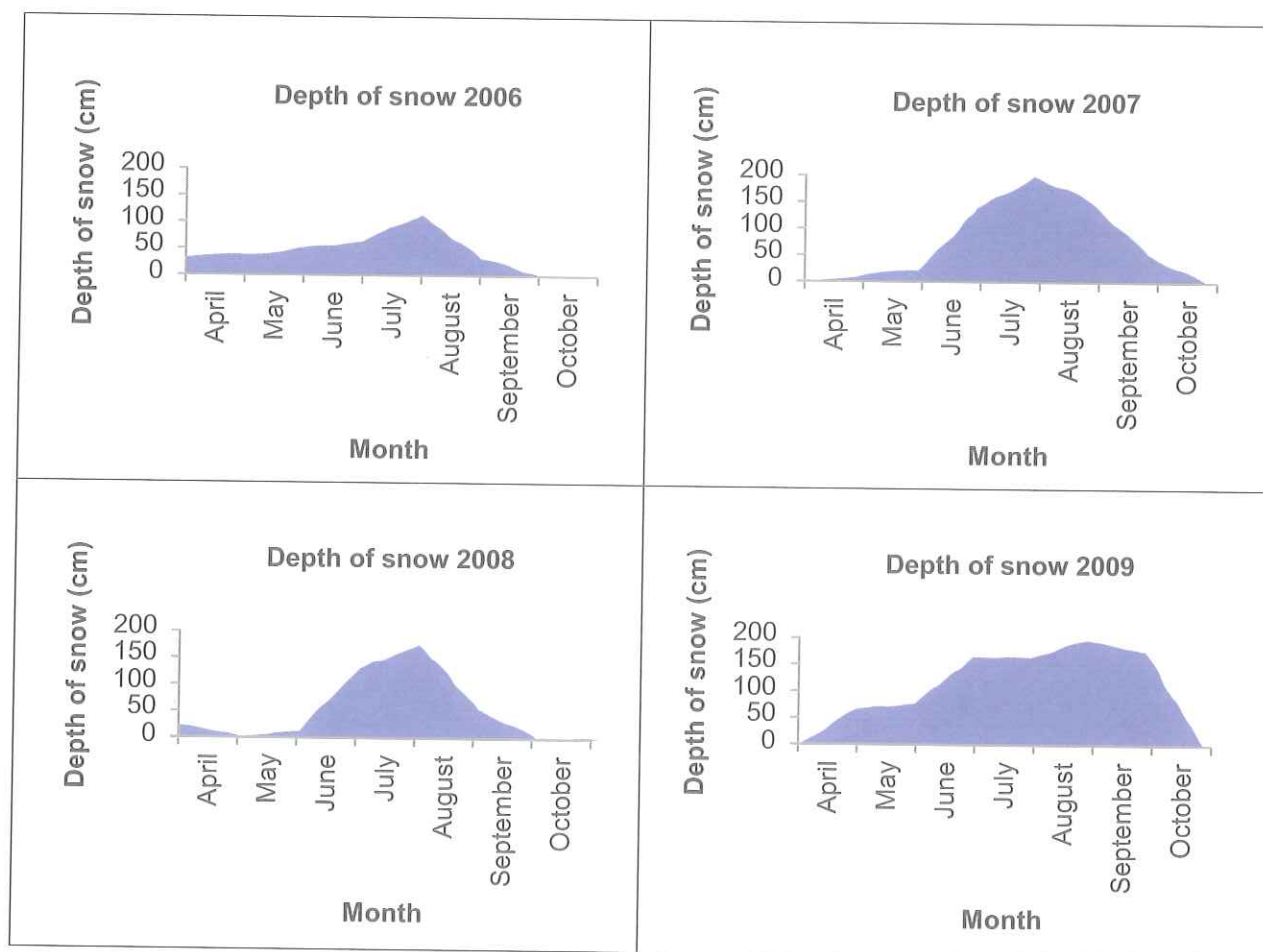
19. Tropical cyclones form only in regions where the ocean water is warm and the air is warm.

Which point on this map shows where tropical cyclones are most likely to form?



20. The depth of snow was measured at the same location from April to October over four years.

These charts show the depth of snow during each of these months over this time.



According to the charts, which year had the greatest total amount of snow from April to October?

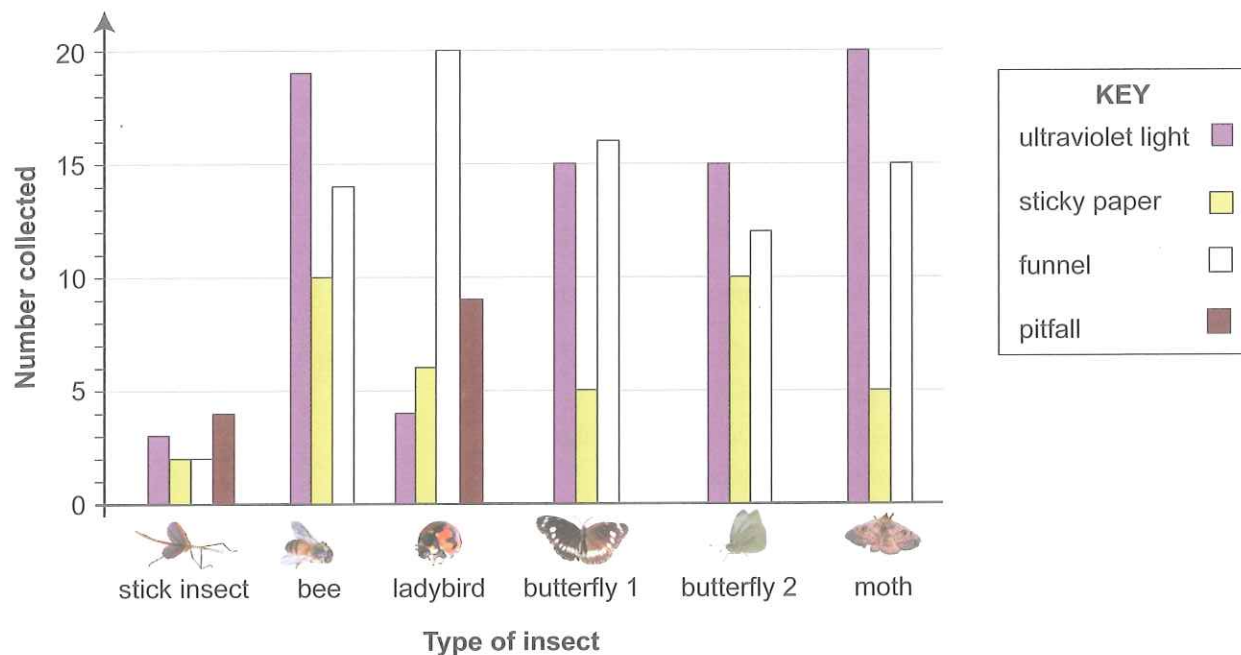
- (A) 2006 (B) 2007 (C) 2008 (D) 2009

For questions 21 and 22 use the information below.

Emily set up four different traps to catch insects:

- an ultraviolet light trap
- a sticky paper trap
- a funnel trap
- a pitfall trap.

The graph shows the number of each type of insect she collected from the traps after 24 hours.



21. Which trap collected the most ladybirds?

- ultraviolet light
- sticky paper
- funnel
- pitfall

22. From the information in Emily's graph, which of the following statements is true?

- The funnel trap caught more bees than moths.
- Bees were best collected using a sticky paper trap.
- Stick insects were the only insects caught in all the traps.
- The pitfall trap was the least effective for collecting insects.

23. The table shows the approximate speed at which light and sound travel in different substances.

Substance	Speed of light (m/s)	Speed of sound (m/s)
air	300 000 000	340
water	225 000 000	1 484
glass	197 000 000	5 960

Based on the information in the table, which statement is correct?

- (A) Light slows down as it passes from water to glass.
(B) Light travels at the same speed through all substances.
(C) Sound speeds up as it passes from glass to water.
(D) Sound travels faster than light in some substances.

24. Ann carried out an experiment to determine how long it took to boil salt water.

The steps of the procedure are listed but they are not in the correct order:

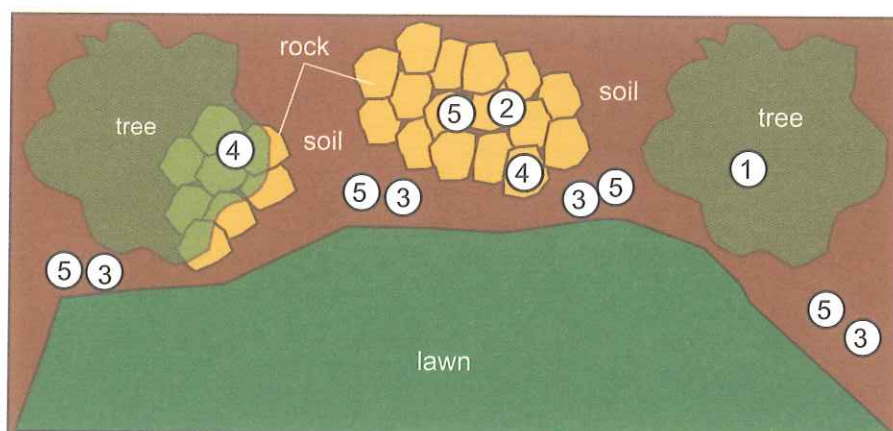
1. Heat the beaker using a Bunsen burner until bubbles form in the water.
2. Weigh 5 grams of salt and put it in the beaker.
3. Measure 150 mL of water and pour it into a beaker.
4. Stop the stopwatch and record the time.
5. Start the stopwatch.

In which order should Ann carry out these steps?

- (A) 3, 2, 5, 4, 1
(B) 3, 2, 5, 1, 4
(C) 5, 3, 1, 2, 4
(D) 5, 2, 3, 4, 1

25. Flowering plants grow best in conditions that suit them. Different plants are suited to different conditions.

The map shows the conditions in a garden that best suit some types of plants.

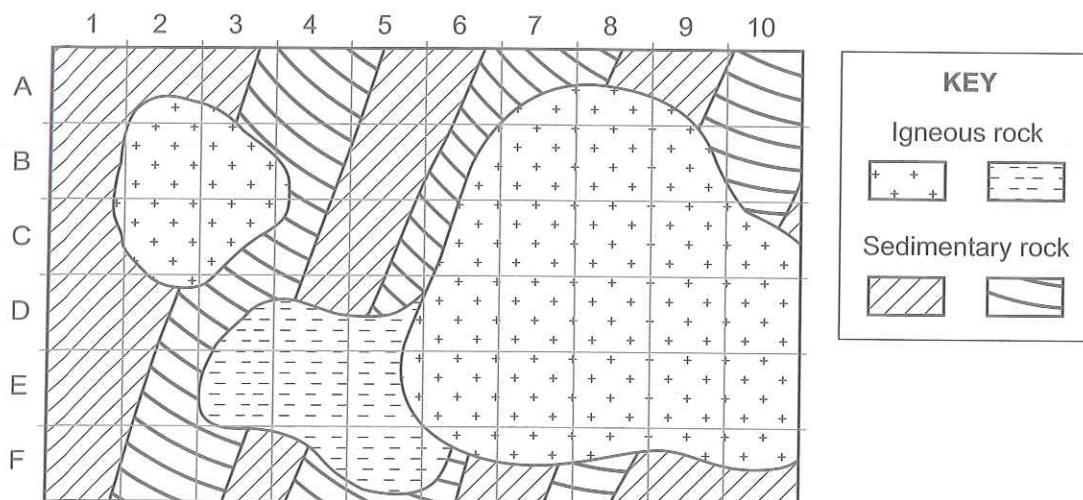


Which plant grows best only in the shade of a tree?

- (A) type 1 (B) type 2 (C) type 3 (D) type 4

26. This map shows the types of rock found in a particular region.

Deposits of quartz are often located at the boundary of igneous and sedimentary rock. Gold is sometimes found in rock containing quartz.

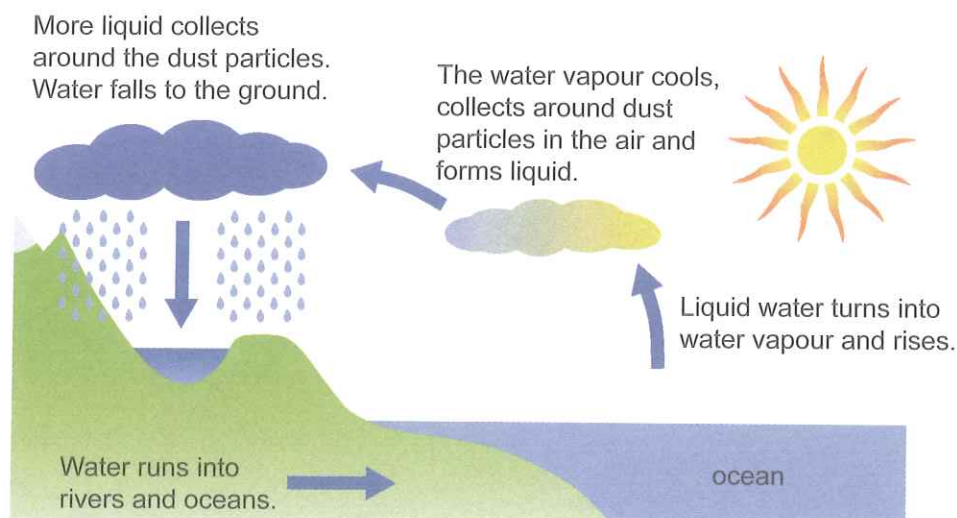


Which of the following locations is most likely to contain gold?

- (A) A6 (B) B2 (C) C3 (D) E5

For questions 27 and 28 use the information below.

This diagram of the water cycle shows how water changes and moves in, on and around the Earth.



27. Which of these supplies the energy for the water cycle?
- (A) Sun (B) rain (C) ocean (D) cloud
28. In areas with low rainfall, it is possible to produce rain by 'seeding' the clouds. Cloud seeding involves dropping particles such as silver iodide into the atmosphere.

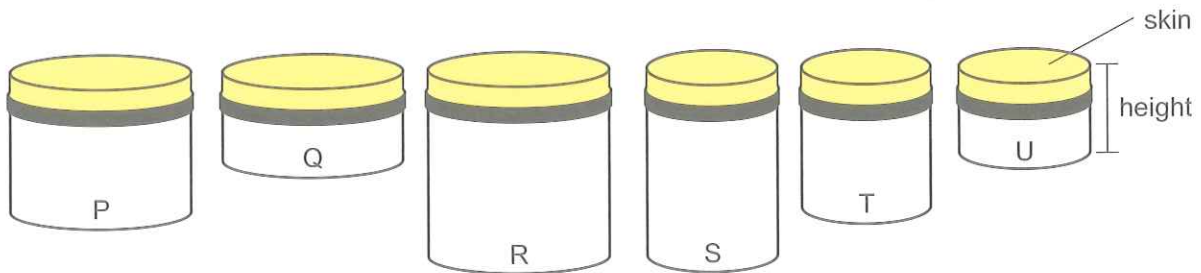
How does the silver iodide increase the amount of rain produced?

- (A) by increasing the amount of water vapour rising from the ocean
- (B) by increasing the number of particles to collect water vapour
- (C) by increasing the speed at which liquid water evaporates
- (D) by increasing the temperature of the atmosphere

29. The pitch of a sound is how high or how low it is.

Julie wants to investigate what happens to the pitch of a drum when she changes the height of the drum, and when she changes the area of the skin at the top of the drum.

The diagram shows some drums.



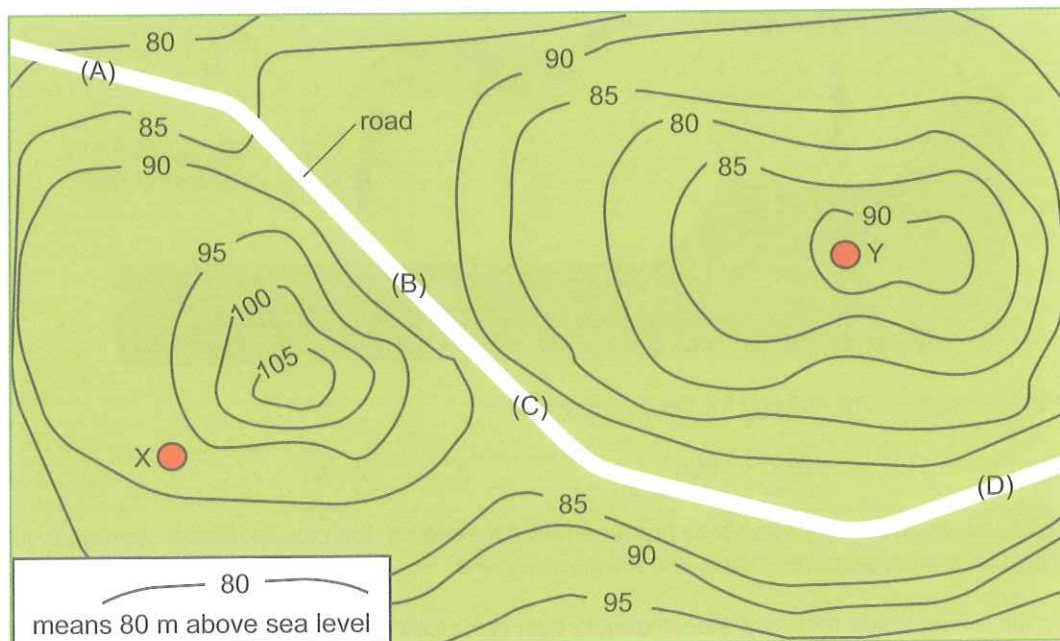
Which drums should Julie use in her investigation?

	Changing the height	Changing the area of the skin
(A)	P and Q	Q and T
(B)	Q and R	P and U
(C)	R and S	P and T
(D)	T and U	Q and U

30. Contour lines on a map join places that are the same height above sea level.

The map shows the contour lines for an area with a road passing through it. There are two towers, each ten metres high. One of the towers is located at X and the other at Y.

From which place along the road is only one of the towers visible?



31. Khalid wanted to find out which of the four laundry liquids he had was better for removing cooking oil stains. He carried out the following steps:
- He dipped a large piece of cloth into cooking oil and then cut the oil-stained cloth into four equal pieces.
 - He mixed each laundry liquid with warm water.
 - He used each mixture to scrub a piece of oil-stained cloth.

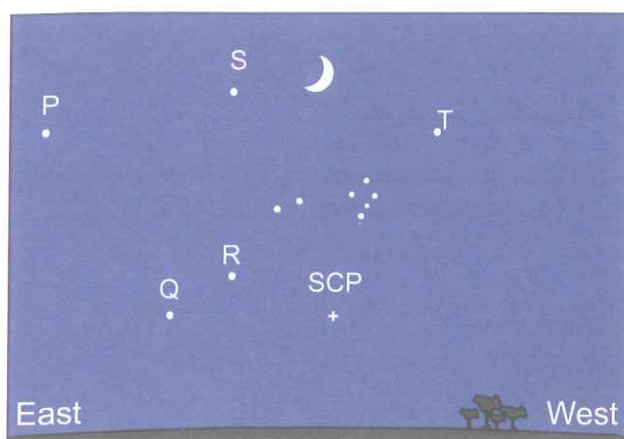
What conditions should Khalid keep the same to have a fair test?

- (A) type of laundry liquid, type of oil, amount of stain left
 (B) water temperature, amount of laundry liquid, scrubbing time
 (C) amount of laundry liquid, scrubbing time, amount of stain left
 (D) water temperature, amount of laundry liquid, type of laundry liquid

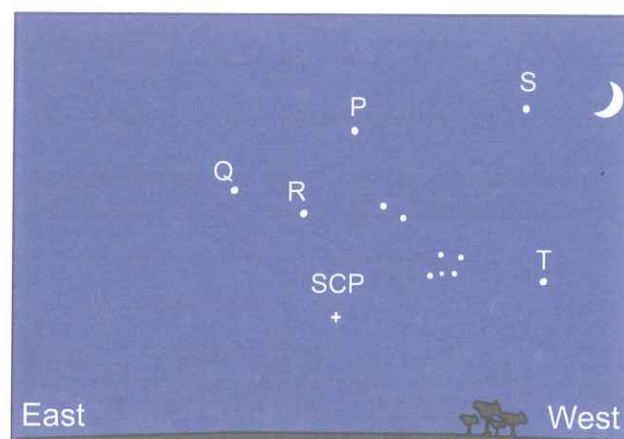
32. When viewing the night sky, looking south from a particular place,

- the Moon and the planets appear to rise in the east and set in the west
- stars appear to revolve around a point called the South Celestial Pole (SCP).

The diagrams show a view of the night sky looking south at two different times on the same evening. The South Celestial Pole (SCP), the Moon, a group of stars and five objects P, Q, R, S and T are shown.



View of sky looking south at 8 pm



View of sky looking south at 11 pm

Which objects are most likely to be planets, and which objects are most likely to be stars?

	Most likely to be planets	Most likely to be stars
(A)	P, S	Q, R, T
(B)	P, S, T	Q, R
(C)	Q, R	P, S, T
(D)	Q, R, T	P, S

33. Different substances have different densities.

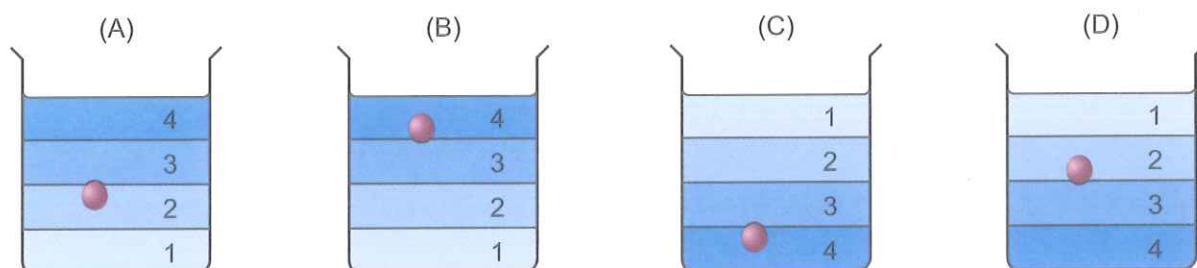
A less dense liquid will float on a liquid that is more dense. A solid will sink in a liquid that is less dense than the solid, and float on a liquid that is more dense.

Ben had four liquids with different densities. The table shows the liquids from least dense to most dense.

	Least dense	←————→			Most dense
Liquid	1	2	3	4	

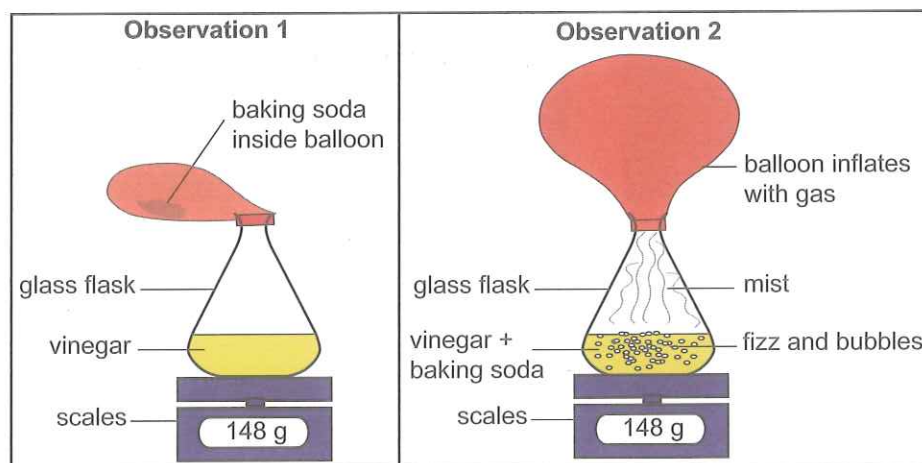
Ben dropped a grape into a beaker containing a layer of each of these liquids. The grape is less dense than liquid 3.

Which beaker shows the result of Ben's experiment?



34. When white baking soda powder is added to vinegar a chemical change occurs.

Nick fitted a balloon containing some baking soda to the top of a glass flask containing vinegar as shown in Observation 1. Observation 2 shows what happened when the baking soda was added to the vinegar.



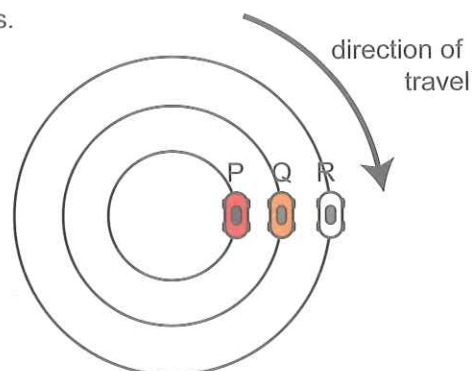
According to Nick's drawings, what shows that a chemical change has taken place, and what was his most likely purpose in using the scales?

	Sign of chemical change	Purpose of the scales
(A)	bubbles, gas forming	to show that the total mass remains the same
(B)	bubbles, gas forming	to measure the amount of gas produced
(C)	vinegar and baking soda	to show that the total mass remains the same
(D)	vinegar and baking soda	to measure the amount of gas produced

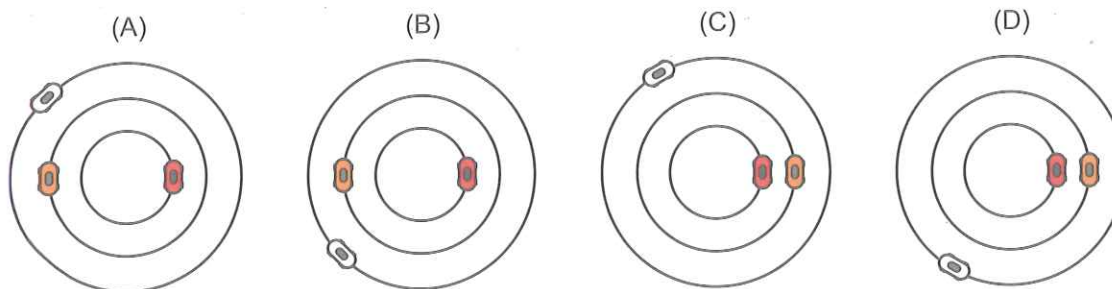
35. Three toy trains P, Q and R are travelling around three separate tracks. All three trains are travelling at the same speed.

Train P takes 1 minute to completely travel around its track.
Train Q takes 2 minutes to completely travel around its track.
Train R takes 3 minutes to completely travel around its track.

At certain times the three trains are level with each other as shown.



Which diagram shows the position of the trains five minutes later?



36. Plants absorb water through their roots. The water rises up through the stem and escapes through tiny holes in the leaves. Vaseline smeared on the leaves stops water from escaping through these tiny holes.

Adrian set up an experiment to find out where these holes are found on leaves.

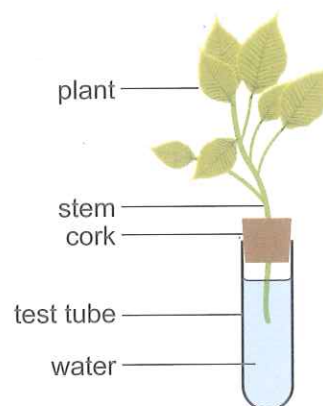
Adrian poured the same amount of water into four test tubes. He placed identical plants in each test tube. He treated some of the leaves with vaseline and left the plants for three days.

The table shows his results.

Treatment	Level of water in test tube (cm)	
	At start	After 3 days
vaseline on top and bottom of leaf	15	13.5
vaseline on top of leaf only	15	11
vaseline on bottom of leaf only	15	12.8
no vaseline on the leaf	15	10.7

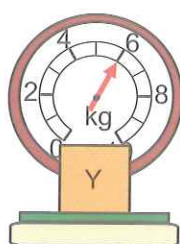
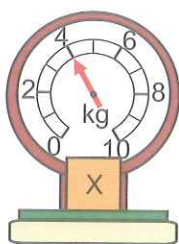
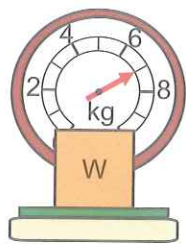
Adrian's results show that water was lost mainly

- (A) through the top of the leaves.
(B) through the bottom of the leaves.
(C) by evaporation from the test tube.
(D) through the stem of the plant.



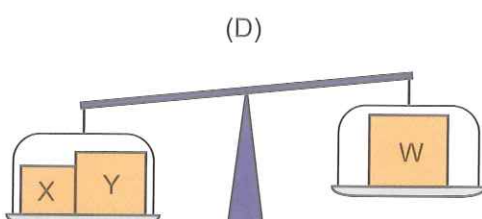
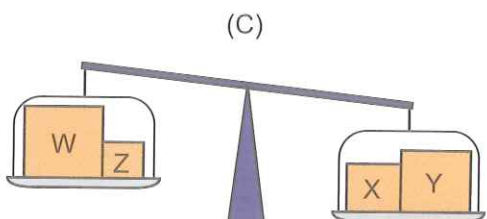
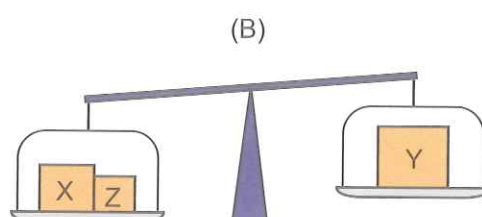
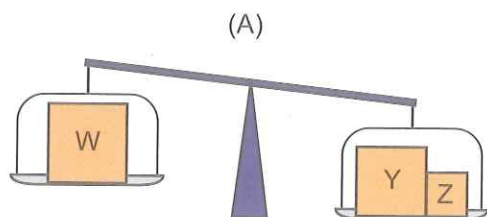
For questions 37 and 38 use the information below.

Frans used scales to find the mass of each of four different objects.



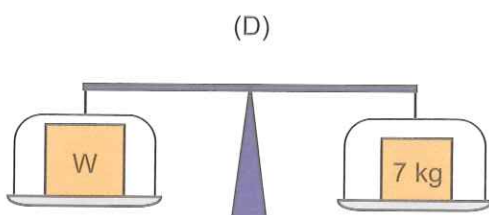
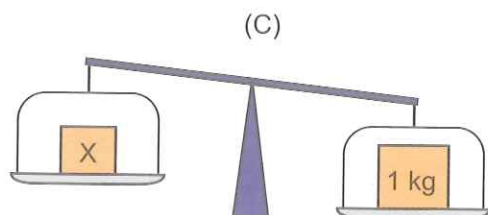
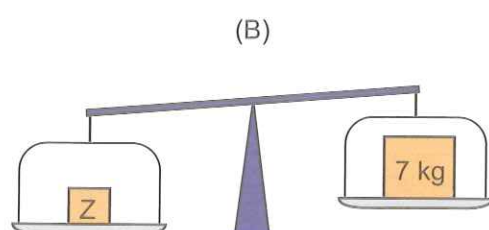
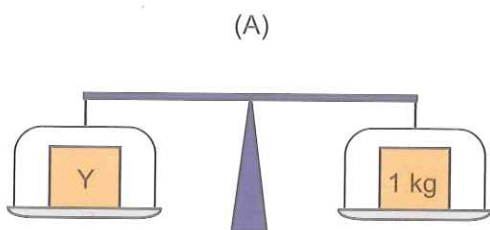
He then compared the masses of the objects using a balance.

37. Which diagram shows the balance comparison that is **NOT** correct?

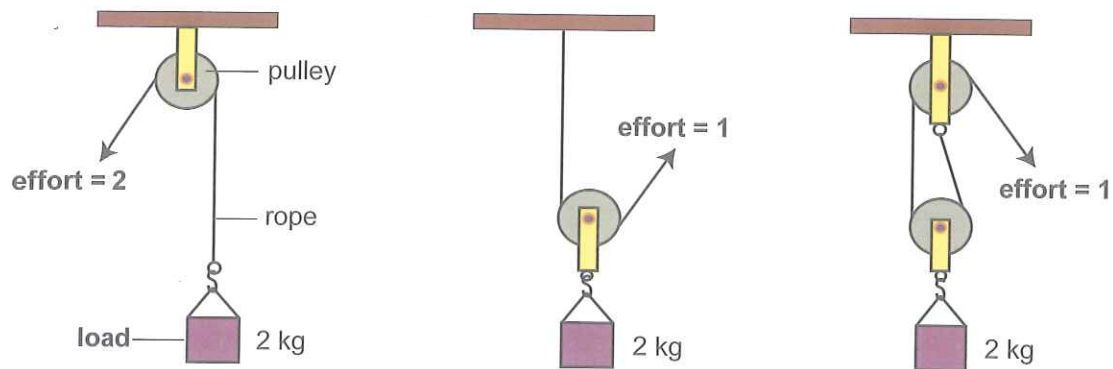


38. On the Moon the force of gravity is one sixth of the force of gravity on Earth. The diagrams show some of the objects placed on a balance on the Moon.

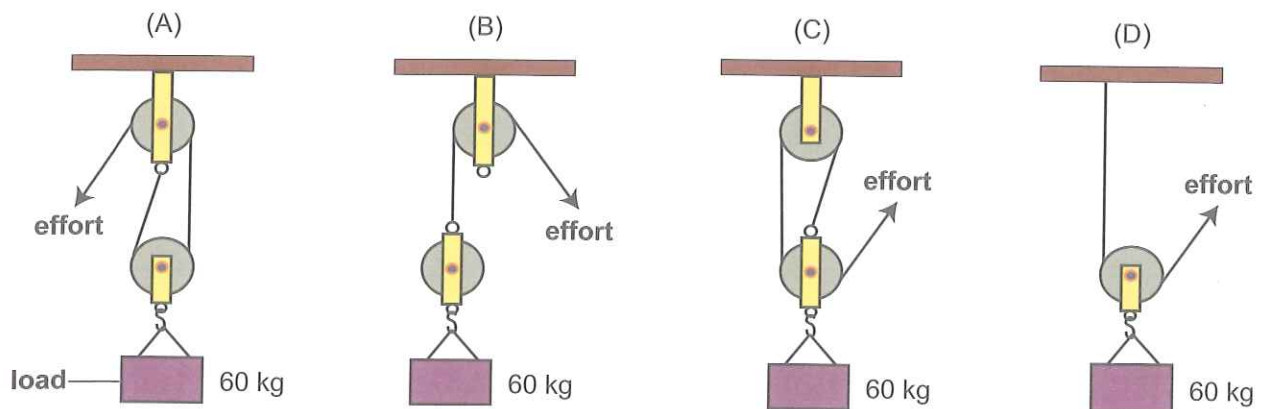
Which of the diagrams is correct?



39. The diagrams show how different pulley systems can be used to reduce the amount of effort required when lifting a load.



Which pulley system shown below would lift a 60 kg load with the least amount of effort?



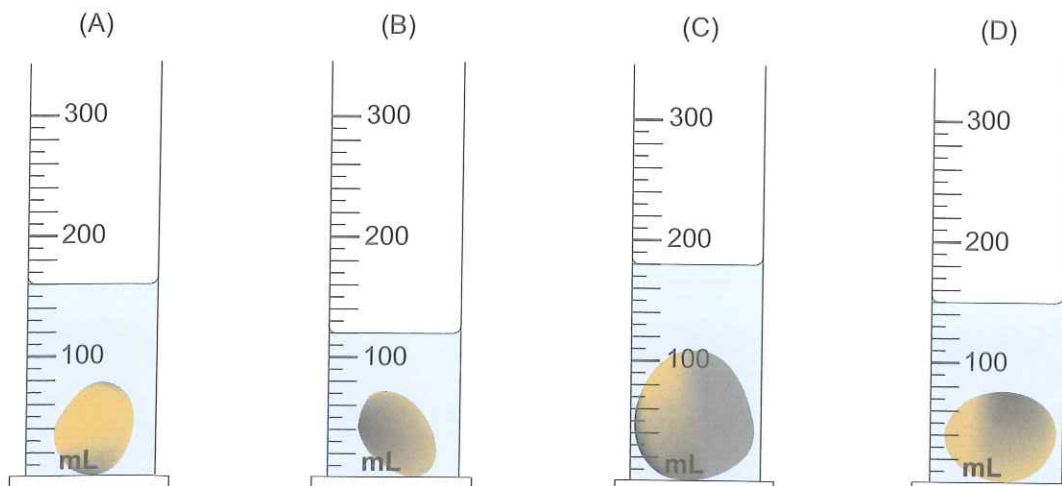
40. The density of an object can be calculated using

$$\text{density} = \text{mass} \div \text{volume}$$

The diagrams show four pebbles of different materials placed in measuring cylinders each of which contained 100 mL of water. Each pebble has a mass of 100 g.

One of the pebbles is composed of the metallic mineral iron pyrites. Iron pyrites has a density of 5 g/mL.

Which cylinder contains the pebble composed of iron pyrites?



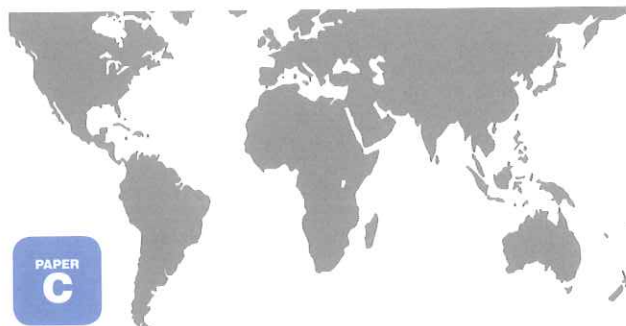
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Indonesia	Year 6
Malaysia	Standard 5
New Zealand	Year 6
Pacific	Year 5
Singapore	Primary 4
South Africa	Grade 5



PAPER
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28	C
29	C
30	B
31	B
32	A
33	D
34	A
35	A
36	A
37	C
38	D
39	D
40	B